Lab 6: Is extreme rainfall at Hobby changing?

2026-10-02

In this lab, you will build your own nonstationary model of extreme daily rainfall at Houston Hobby Airport and decide whether the evidence supports a published claim that it is rising. Everyone uses the same record, and each of you makes different choices along the way. On Wednesday, October 7, you will present your results in class and defend those choices, as practice for the oral exam.

Objectives

By the end of this lab, you will be able to:

1. Build a nonstationary extreme value model, choosing annual maxima or peaks over threshold, a covariate, and the form of the trend.
2. Report a change in the 100-year rainfall with an interval, and say what the interval leaves out.
3. Defend each choice behind your result.

Setup

First, load the packages.

```
using Pkg
here = dirname(@__FILE__)
Pkg.activate(here)
Pkg.instantiate()

using CEVE543Utils
using CairoMakie, CSV, DataFrames, Dates, Distributions, LaTeXStrings
using Printf, Random, Statistics
using FlexiChains: summarystats

rng = MersenneTwister(543)
set_theme!(theme_minimal(); fontsize=14)
colors = Makie.wong_colors()
```

Rainfall data

You will analyze daily precipitation at Houston Hobby Airport, station `USW00012918` in NOAA's Global Historical Climatology Network, which is the record from Monday's lectures. The file `hobby-daily.csv` has one row for each day with a valid reading, with the rainfall in the column `prcp_mm`, in millimeters.

Figure 1 plots every wet day.

```
daily = CSV.read(joinpath(here, "hobby-daily.csv"), DataFrame)
daily.year = year.(daily.date)
wet = filter(r -> r.prcp_mm > 0, daily)

fig = Figure(; size=(700, 300))
ax = Axis(fig[1, 1]; xlabel=L"\text{date}", ylabel=L"\text{daily rainfall (mm)}")
scatter!(ax, wet.date, wet.prcp_mm; color=colors[1], markersize=3)
fig
```

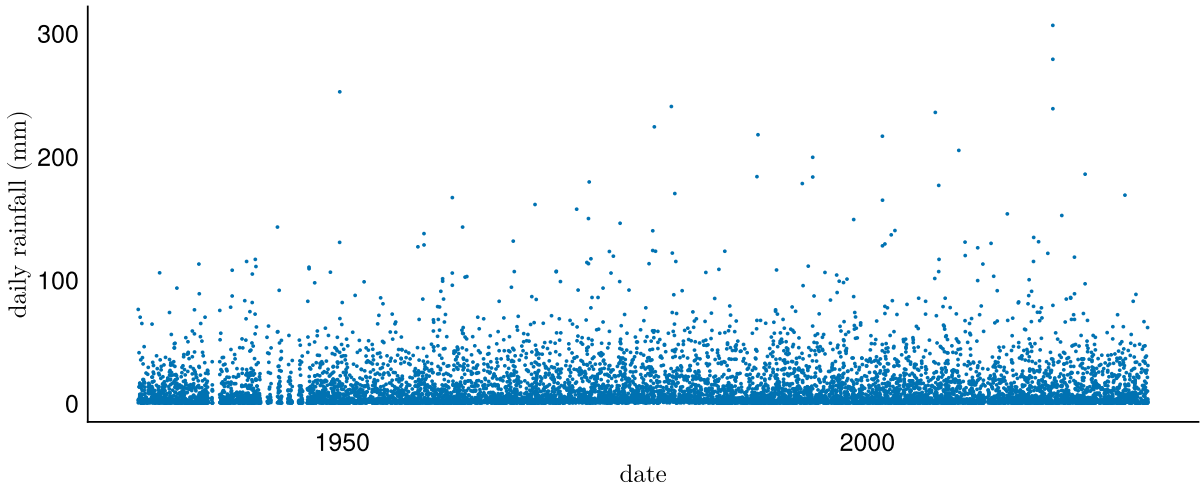


Figure 1: Daily rainfall at Houston Hobby Airport, wet days only.

Not every year is complete. Figure 2 counts the days with a valid reading in each year.

```
days_per_year = combine(groupby(daily, :year), nrow => :n_days)
fig = Figure(; size=(700, 300))
ax = Axis(fig[1, 1]; xlabel=L"\text{year}", ylabel=L"\text{days with a valid reading}")
scatterlines!(ax,    days_per_year.year,    days_per_year.n_days;    color=colors[1],
markersize=5)
fig
```

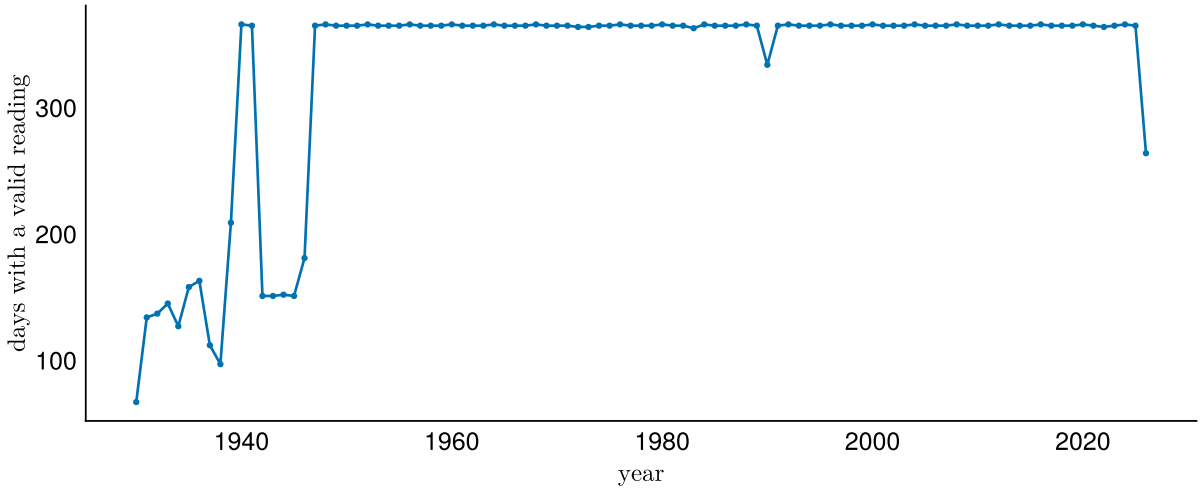


Figure 2: Days with a valid reading in each year at Houston Hobby Airport.

Many years are missing days, and 2026 is not over yet, so you need to decide how to handle the gaps. Assume the data are *missing at random*: whether a day was recorded has nothing to do with how much rain fell. To see when this might fail, picture a rain gauge that breaks in the heaviest storms. A simple approach is to keep only the years with at least some number of valid days, such as 350. You are welcome to try other cutoffs or other ways to filter.

```
keep = days_per_year.year[(days_per_year.n_days .>= 350) .& (days_per_year.year .<=
2025)]
filter!(r -> r.year in keep, daily)
```

Covariates

You have three candidate covariates in `covariates.csv`, one row per year from 1880 to 2025:

Column	What it is	Source
`log_CO2`	Natural log of annual mean atmospheric CO2, in parts per million	Before 1959, the Law Dome ice core record [1], [2]; from 1959, the Mauna Loa annual means from NOAA's Global Monitoring Laboratory
`gmst`	Global mean surface temperature anomaly, in °C relative to 1951-1980	NASA's GISTEMP version 4, January to December mean
`nino34`	Sea surface temperature anomaly in the Niño 3.4 region of the tropical Pacific, in °C relative to 1981-2010	HadISST, from NOAA's Physical Sciences Laboratory, mean of the 12 calendar months

CO2 and global temperature both rise over the record, so they are candidates for a long-term trend. Niño 3.4 measures the El Niño-Southern Oscillation, which shifts from year to year with no long-term trend, so it is a candidate for explaining which years are wet.

```
covariates = filter(r -> r.year >= 1930, CSV.read(joinpath(here, "covariates.csv"),
DataFrame))
fig = Figure(; size=(700, 500))
for (k, (col, label)) in enumerate(((:log_CO2, "log CO2"), (:gmst, "GMST anomaly (°C)"),
(:nino34, "Niño 3.4 anomaly (°C)")))
    lines!(Axis(fig[k, 1]; ylabel=label, xlabel=k == 3 ? "year" : ""), covariates.year,
covariates[!, col]; color=colors[k])
end
fig
```

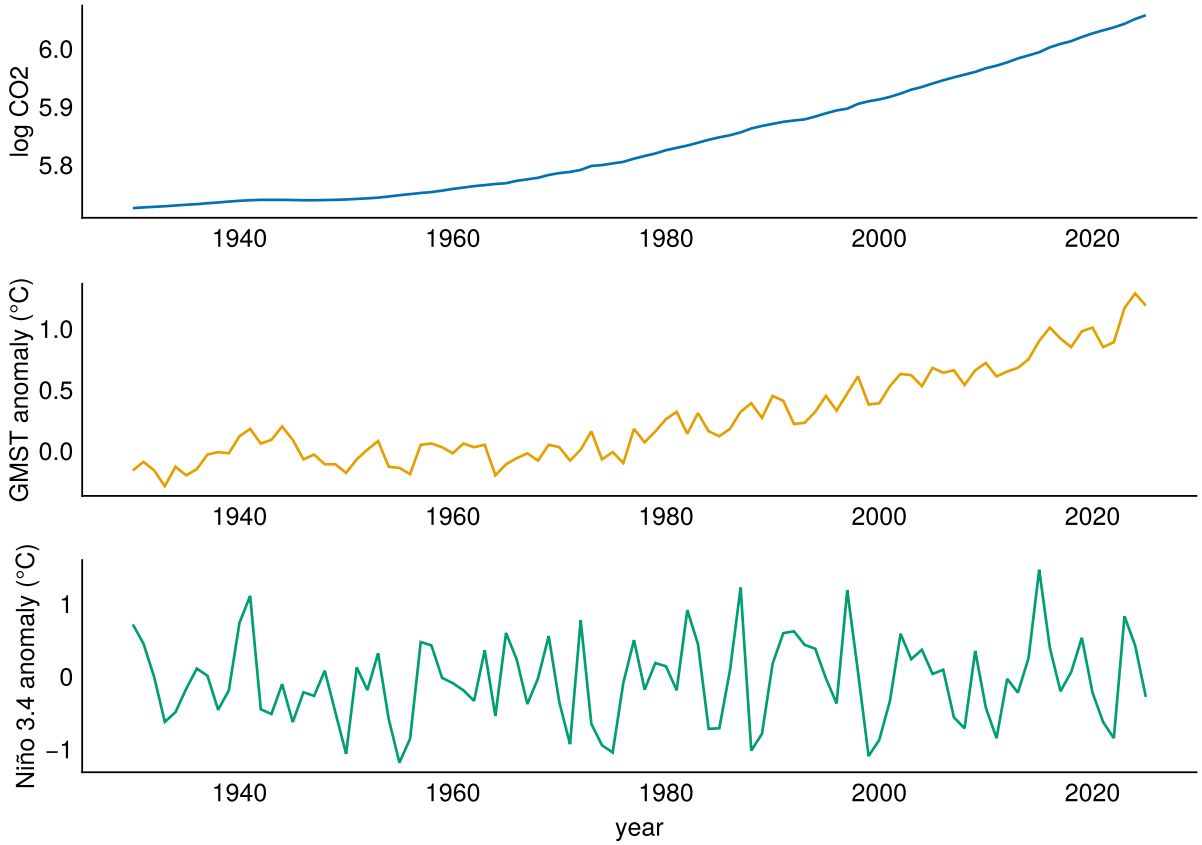


Figure 3: The three candidate covariates over the years of the Hobby record.

Your challenge

C. Fagnant, A. Gori, A. Sebastian, P. B. Bedient, and K. B. Ensor [3] analyzed more than 600 rain gauges in southeast Texas and reported that the 100-year daily rainfall at Hobby rose by 1.93 mm per year (95% interval 1.50 to 2.35). In this lab, you will analyze the same station, choose your own rule for which years to keep, and fit your own model. Their rule for keeping years differs from yours, so your answer need not

match theirs. You will choose whether to fit a GEV to annual maxima or a GPD to peaks over threshold, which covariates to use and how each parameter depends on them, and which priors to include.

Prepare a 5-10 minute slide presentation for class on Wednesday, October 7, that leads with your results and then explains your modeling choices. Upload your slides to the Canvas page for the Lab 6 presentations before class on Wednesday. Answer these four key questions.

1. What is your best estimate for the 10- and 100-year return levels in 2026?
2. How uncertain are these estimates?
3. How have these return levels changed over the past 40 years?
4. How will these return levels change over the next 40 years, and what assumptions do you make in projecting these?

Explain each key decision below, saying what you chose and why.

1. Did you fit a GEV to annual maxima or a GPD to peaks over threshold? If you fit a GEV, how did you handle years with missing data? If you fit a GPD, how did you decluster, and how did you choose the threshold?
2. How did you decide between a stationary and a nonstationary model? If nonstationary, how did you choose the covariates and the functional form? If stationary, what gave you confidence in that choice?
3. How did you estimate your uncertainty? If you used a Bayesian fit, how did you choose the priors? If you used a bootstrap, how did you choose how to resample?

Examples

You make more of the choices in this lab than in any earlier one. Two worked examples on synthetic (simulated) data give you all the code you need. Keep this section as a reference while you work; you may delete all of it, both the GEV and the GPD examples, from the notebook you turn in.

GEV example

In the first example, you fit a GEV with a quadratic trend in the location and the log scale. A quadratic is a poor choice for a real record, for the reasons under Other functional forms, but it uses every keyword you will need.

$$y_t \sim \text{GEV}\left(\alpha_0 + \alpha_1 x_t + \alpha_2 x_t^2, \exp(\beta_0 + \beta_1 x_t + \beta_2 x_t^2), \xi\right).$$

The covariate is the year centered on 1985, $x_t = t - 1985$. The synthetic dataset has one annual maximum for each year from 1946 to 2025. In 1985 its location rises by 0.5 mm per year and its scale shrinks by 0.5% per year; by 1990 both trends have reversed.

```
years = collect(1946:2025)
x = years .- 1985
μ_true = 90 .+ 0.5 .* x .- 0.05 .* x .^ 2
σ_true = 35 .* exp.(-0.005 .* x .+ 0.0005 .* x .^ 2)
y_syn = rand.(rng, GeneralizedExtremeValue.(μ_true, σ_true, 0.15))
```

The `gevfitbayes` function from the `CEVE543Utils` package fits a GEV whose location and log scale are each a linear combination of the covariates you pass. Passing x_t and x_t^2 as two covariates gives the quadratic above:

```
gev_syn = gevfitbayes(y_syn; locationcov=[x, x .^ 2], logscalecov=[x, x .^ 2], rng=rng,
progress=false)
DataFrame(summarystats(gev_syn.estimate))
```

```
┌ Warning: failed to find valid initial parameters in 10 tries; consider providing a
different initialisation strategy with the `initial_params` keyword
└ @ Turing.Inference ~/.julia/packages/Turing/4sXd9/src/mcmc/abstractmcmc.jl:63
┌ Warning: failed to find valid initial parameters in 10 tries; consider providing a
different initialisation strategy with the `initial_params` keyword
└ @ Turing.Inference ~/.julia/packages/Turing/4sXd9/src/mcmc/abstractmcmc.jl:63
┌ Warning: failed to find valid initial parameters in 10 tries; consider providing a
different initialisation strategy with the `initial_params` keyword
└ @ Turing.Inference ~/.julia/packages/Turing/4sXd9/src/mcmc/abstractmcmc.jl:63
┌ Warning: the tail effective sample size is 24% of the draws, below 25%, so the draws
are far from independent. Keep every 5th draw with `thin(fit, 5)`, or sample again with
`thinning=5` to keep `n_samples` draws per chain; if the share stays low the chains
may be stuck, so try `sampler=:demczs`.
└ @ CEVE543Utils ~/.julia/packages/CEVE543Utils/wB1FE/src/fitting.jl:351
```

Row	param	mean	std	mcse	ess_bulk	ess_tail	rhat	q5	q50	q95
	VarName	Float64	Float64	Float64	Float64	Float64	Float64	Float64	Float64	Float64
1	β_location[1]	67.0963	5.84589	0.204974	815.438	1446.22	1.00152	57.4133	66.9317	76.7413
2	β_location[2]	16.5278	6.09354	0.199623	935.086	1294.99	1.00338	6.21113	16.5231	26.6017
3	β_location[3]	−21.3601	6.01303	0.203295	875.332	1616.49	1.00526	−31.1275	−21.4385	−11.4268
4	β_logscale[1]	3.74746	0.100699	0.00346467	848.234	1225.67	1.00099	3.58602	3.74687	3.91614
5	β_logscale[2]	0.122330	0.0968260	0.00340747	806.303	1245.12	1.00256	−0.180694	0.0257387	0.139678
6	β_logscale[3]	−0.0127	0.109830	0.00413586	707.9	1207.26	1.00459	0.0274186	0.201457	0.392385
7	β_shape[1]	0.289429	0.110946	0.0039076	822.978	964.21	1.00108	−0.145494	0.0235015	0.218836

`locationcov` and `logscalecov` each take a list of covariates, so `[x, x .^ 2]` is two of them. `rng` is the random number generator from the setup cell, and `progress=false` hides the progress bar.

By default, priors in `gevfitbayes` are flat on every coefficient and uniform between -0.5 and 0.5 on ξ. The posterior is sampled with DE-MCzs, the sampler in the RMC-BestFit software that C. H. Smith, B. Skahill, and D. A. Margo [4] use. Pass `sampler=NUTS()` to use the NUTS sampler instead.

The `summarystats` table is for checking the chains. Check the `rhat` column, which should be close to 1, and the `ess_bulk` column, which should be in the hundreds or more, as in lab 5. DE-MCzs takes no gradient steps, so it has no divergences to count. Before sampling, the fit centers each covariate and divides it by its standard deviation, so a slope in that table is a change per standard deviation of the covariate.

`params` gives the coefficients in the units of the covariate: a matrix for each of `β_location`, `β_logscale`, and `β_shape`, with one row per posterior draw and one column per coefficient. The first column is the

intercept, which is the value at the means of the covariates rather than at $x_t = 0$. The second column is α_1, the slope per year in 1985, and the third is α_2.

```
coefs = params(gev_syn)
α1 = coefs.β_location[:, 2]
@printf("α1: median %.2f mm per year [%.2f, %.2f]; the true value is 0.5\n",
    median(α1), quantile(α1, 0.025), quantile(α1, 0.975))
```

```
α1: median 0.71 mm per year [0.19, 1.22]; the true value is 0.5
```

`returnlevel` computes return levels at any covariate value you pass, giving one row per posterior draw and one column per value. Here the values are the years you want minus 1985, and their squares. So `rl100[:, 1]` is the 100-year level in 1986, `rl100[:, 2]` in 2026, and `rl100[:, 3]` in 2066, with one entry per draw. The cell also defines `interval`, which rounds the 2.5%, 50%, and 97.5% quantiles of the draws to whole millimeters: a 95% interval with the median in the middle. Each `println` line reports the 100-year level; to report the 10-year level, use `rl10` in place of `rl100`.

```
report_years = [1986, 2026, 2066]
x_report = report_years .- 1985
rl10   =   returnlevel(gev_syn,   10;   locationcov=[x_report,   x_report   .^   2],
logscalecov=[x_report, x_report .^ 2])
rl100   =   returnlevel(gev_syn,   100;   locationcov=[x_report,   x_report   .^   2],
logscalecov=[x_report, x_report .^ 2])

interval(v) = round.(quantile(v, [0.025, 0.5, 0.975]))
println("100-year level in 2026 (mm): ", interval(rl100[:, 2]))
println("Change, 1986 to 2026 (mm): ", interval(rl100[:, 2] .- rl100[:, 1]))
println("Change, 2026 to 2066 (mm): ", interval(rl100[:, 3] .- rl100[:, 2]))
```

```
100-year level in 2026 (mm): [202.0, 369.0, 852.0]
Change, 1986 to 2026 (mm): [-79.0, 107.0, 566.0]
Change, 2026 to 2066 (mm): [-221.0, 1895.0, 53377.0]
```

The projection to 2066 assumes the quadratic keeps going for another 40 years, past the end of the data. With log CO2 or GMST as the covariate, you also have to choose a value for that covariate in 2066, and say what it is and why.

Figure 4 plots the fitted 2- and 100-year levels in each year of the record next to the true levels the synthetic data came from. With no covariates passed, `returnlevel` evaluates at each year in the record.

```
fig = Figure(; size=(700, 350))
ax = Axis(fig[1, 1]; xlabel=L"\text{year}", ylabel=L"\text{daily rainfall (mm)}")

for (k, T) in enumerate([2, 100])
    rl_fit = returnlevel(gev_syn, T)
```

```
    rl_true = returnlevel.(GeneralizedExtremeValue.(μ_true, σ_true, 0.15), T)
   lines!(ax, years, median.(eachcol(rl_fit)); color=colors[k], linewidth=2, label="$T-
year level, fitted")
    lines!(ax, years, rl_true; color=colors[k], linestyle=:dash, label="$T-year level,
true")
end

scatter!(ax, years, y_syn; color=:black, markersize=5, label="synthetic annual maxima")
axislegend(ax; position=:lt)
fig
```

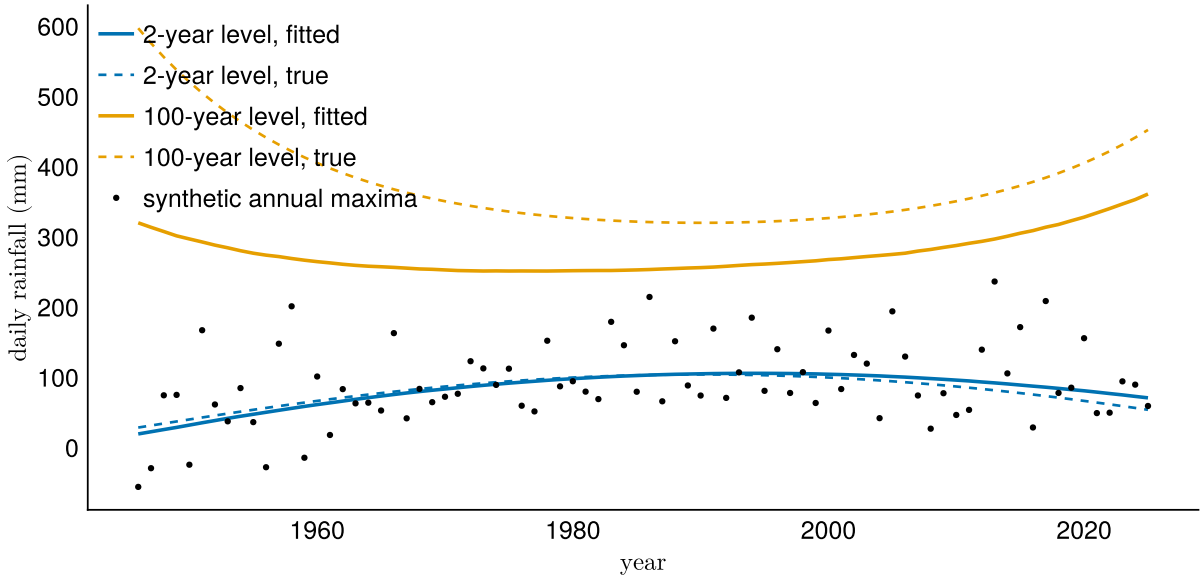


Figure 4: The 2- and 100-year levels through time from the GEV fit to the synthetic record (posterior median), the true levels, and the synthetic annual maxima.

You can also fit by maximum likelihood and get an interval from a bootstrap. `gevfit` fits the same model by maximum likelihood, and `getdistribution` gives the fitted GEV for each year. A parametric bootstrap draws a new record from those distributions, refits it, and records its return levels, many times over. The spread of the refitted levels is the uncertainty.

```
gev_mle = gevfit(y_syn; locationcov=[x, x .^ 2], logscalecov=[x, x .^ 2])
gev_by_year = getdistribution(gev_mle)

n_boot = 500
rl100_boot = zeros(n_boot, length(report_years))
for b in 1:n_boot
    y_boot = rand.(rng, gev_by_year)
    fit_boot = gevfit(y_boot; locationcov=[x, x .^ 2], logscalecov=[x, x .^ 2])
    rl100_boot[b, :] = returnlevel(fit_boot, 100; locationcov=[x_report, x_report .^
2], logscalecov=[x_report, x_report .^ 2])
```

```
end

println("100-year level in 2026 (mm): ", interval(rl100_boot[:, 2]))
println("Change, 1986 to 2026 (mm): ", interval(rl100_boot[:, 2] .- rl100_boot[:, 1]))
println("Change, 2026 to 2066 (mm): ", interval(rl100_boot[:, 3] .- rl100_boot[:, 2]))
```

```
100-year level in 2026 (mm): [174.0, 311.0, 570.0]
Change, 1986 to 2026 (mm): [-91.0, 70.0, 303.0]
Change, 2026 to 2066 (mm): [-246.0, 977.0, 14336.0]
```

Other functional forms

You choose the functional form: which parameters change with the covariate, and how. Every fitting function also takes a table and the names of its columns. For example, `gevfitbayes(annmax, :y; locationcovid=[:log_CO2], logscalecovid=[:log_CO2])` fits the column `y` of the table `annmax`, with the location and the log scale each a line in `log_CO2`. You build `annmax` under Your Hobby analysis below. With this table form, each functional form is a choice of covariate columns:

- For a trend in the location alone, pass `locationcovid` and leave out `logscalecovid`.
- For a trend in the scale alone, pass `logscalecovid` and leave out `locationcovid`.
- Leave out both to fit a stationary model.
- For a curved trend, add a column holding the covariate squared to your table, such as `annmax.log_CO2_sq = annmax.log_CO2 .^ 2`, and name it in `locationcovid`. A quadratic will find some curvature in almost any record, including one drawn from a straight line. Outside the data it grows without limit, so its 2066 level depends on a curvature the record barely pins down.
- For an abrupt change at a year you choose, add a column that is 0 before that year and 1 after it. Say why that year, and whether you chose it before or after looking at the data.

Prior information

You change the priors with two keywords. `trend_priors` puts a prior on the slopes, one for each parameter, and any you leave out stay flat. The slope priors are on the standardized scale of the `summarystats` table, so each is a change per standard deviation of the covariate. A `location` slope is in millimeters, so `trend_priors=(location=Normal(0, 5),)` says the location probably moves less than about 10 mm per standard deviation. The trailing comma makes it a named tuple with one entry. A `logscale` slope is a proportional change in the scale, so `logscale=Normal(0, 0.1)` says the scale probably changes by less than about 20% per standard deviation. `quantile_priors` states what you believe about return levels, as in lab 4. Each belief gives the position of a year in the record (`index=1` is the first year), a return period, and a distribution for that return level in that year. You can give as many as you like. The two beliefs below are hypothetical, made up to suit the synthetic record. For Hobby, base a belief on a source you can name, and say what it is.

```
beliefs = [
    (index=1, period=100, belief=Normal(250, 40)),
    (index=lastindex(y_syn), period=100, belief=Normal(450, 40)),
]
```

```
gev_beliefs = gevfitbayes(y_syn; locationcov=[x, x .^ 2], logscalecov=[x, x .^ 2],
quantile_priors=beliefs, rng=rng, progress=false)
rl100_beliefs = returnlevel(gev_beliefs, 100; locationcov=[x_report, x_report .^ 2],
logscalecov=[x_report, x_report .^ 2])
@printf("2026 100-year level: median %.0f mm without beliefs, %.0f mm with them\n",
median(rl100[:, 2]), median(rl100_beliefs[:, 2]))
```

```
┌ Warning: failed to find valid initial parameters in 10 tries; consider providing a
different initialisation strategy with the `initial_params` keyword
└ @ Turing.Inference ~/.julia/packages/Turing/4sXd9/src/mcmc/abstractmcmc.jl:63
┌ Warning: failed to find valid initial parameters in 10 tries; consider providing a
different initialisation strategy with the `initial_params` keyword
└ @ Turing.Inference ~/.julia/packages/Turing/4sXd9/src/mcmc/abstractmcmc.jl:63
2026 100-year level: median 369 mm without beliefs, 442 mm with them
```

GPD example

In the second example, you fit a GPD with a trend in the log scale, using the same centered year as the covariate. The synthetic storm record has 12 storms a year above a threshold of 25.4 mm, and the scale of each storm's excess grows by 0.4% per year.

```
x_storm = repeat(x; inner=12)
peaks_syn = 25.4 .+ rand.(rng, GeneralizedPareto.(0, 20 .* exp.(0.004 .* x_storm),
0.15))
```

`gpfitbayes` takes the peaks and the threshold u and fits a GPD to the excesses. Its log scale is a straight line in the covariate, where x_i is the year of storm i minus 1985:

$$y_i - u \sim \mathrm{GPD}\ (\exp(\beta_0 + \beta_1 x_i), \xi).$$

```
gp_syn = gpfitbayes(peaks_syn, 25.4; logscalecov=[x_storm], rng=rng, progress=false)
DataFrame(summarystats(gp_syn.estimate))
```

Row	param	mean	std	mcse	ess_bulk	ess_tail	rhat	q5	q50	q95
	VarName	Float64	Float64	Float64	Float64	Float64	Float64	Float64	Float64	Float64
1	β_logscale[1]	3.03393	0.0491283	0.000949048	2684.82	3362.16	0.999814	2.95298	3.03465	3.11506
2	β_logscale[2]	0.12723	0.0367888	0.000703159	2739.94	3375.66	0.999879	0.067518	0.125911	0.187222
3	β_shape[1]	0.140575	0.0383701	0.000795451	2340.34	3021.72	1.00033	0.0797338	0.139521	0.206701

A GPD has no location parameter to trend, so you pass only `logscalecov`. A GPD return level also needs `rate`, the average number of storms per year above the threshold, which is 12 here. `gpfit` and `gpfitbayes` treat `rate` as one known number: they do not allow a trend in how often storms exceed the threshold, and their return-level intervals leave out the uncertainty in `rate`. If you want to model either, write the model in Turing directly.

```
rl100_gp = returnlevel(gp_syn, 100; rate=12, logscalecov=[x_report])
println("100-year level in 2026 (mm): ", interval(rl100_gp[:, 2]))
println("Change, 1986 to 2026 (mm): ", interval(rl100_gp[:, 2] .- rl100_gp[:, 1]))
```

```
100-year level in 2026 (mm): [266.0, 339.0, 457.0]
Change, 1986 to 2026 (mm): [25.0, 61.0, 116.0]
```

With 960 storms in place of 80 annual maxima, the intervals are narrower.

Your Hobby analysis

Now use the code in the examples to answer the key questions above. If you fit a GEV, this cell makes a table of annual maxima joined to the covariates:

```
annmax = combine(groupby(daily, :year), :prcp_mm => maximum => :y)
annmax = sort(innerjoin(annmax, covariates; on=:year), :year)
```

If you fit a GPD, this cell finds the peaks over a threshold and joins each peak's year to the covariates. The threshold of 25.4 mm and the 72-hour gap between storms are made up to show the code; choose your own and defend them, as in lab 5. `decluster` keeps the largest day in each storm, and `rate` is the number of peaks per year you kept, which you pass to `returnlevel`.

```
u = 25.4
peak_idx, _ = decluster(DateTime.(daily.date), daily.prcp_mm, u; min_gap_hours=72)
peaks = innerjoin(daily[peak_idx, [:year, :prcp_mm]], covariates; on=:year)
rate = nrow(peaks) / length(keep)
```

Fit the peaks with the table form, such as `gpfitbayes(peaks, :prcp_mm, u; logscalecovid=[:log_CO2], rng=rng, progress=false)`.

Turning it in

This lab is due on Canvas on Friday, October 9, one week after the Friday session that starts it.

1. Check that your notebook answers each key question and explains each key decision.
2. Render the notebook to a PDF. In the terminal, from the lab folder, run:

   ```
   quarto render index.qmd --to typst
   ```

   This writes `index.pdf` beside the notebook. It runs the whole notebook from the top in a fresh process, so it also confirms your work does not depend on leftover REPL state.
3. Commit and push your work, with GitHub Desktop or by asking Claude to commit and push.
4. Upload `index.pdf` to the Lab 6 assignment on Canvas.
5. Paste the link to your repository in the same Canvas submission.

If the render fails and you cannot fix it, upload what you have and say where it broke. A lab that does not render is worth more to me than a lab you did not turn in.

Note what broke, what Claude got wrong, and what you had to fix by hand, and bring it to the next session.

Bibliography

[1] D. M. Etheridge, L. P. Steele, R. L. Langenfelds, R. J. Francey, J. Barnola, and V. I. Morgan, "Natural and Anthropogenic Changes in Atmospheric CO\textsubscript{2} over the Last 1000 Years from Air in Antarctic Ice and Firn," *Journal of Geophysical Research: Atmospheres*, vol. 101, no. D2, pp. 4115–4128, Feb. 1996, doi: 10.1029/95jd03410.

[2] C. MacFarling Meure *et al.*, "Law Dome CO \textsubscript{2}, CH\textsubscript{4} and N\textsubscript{2}O Ice Core Records Extended to 2000 Years BP," *Geophysical Research Letters*, vol. 33, no. 14, Jul. 2006, doi: 10.1029/2006gl026152.

[3] C. Fagnant, A. Gori, A. Sebastian, P. B. Bedient, and K. B. Ensor, "Characterizing Spatiotemporal Trends in Extreme Precipitation in Southeast Texas," *Natural Hazards*, vol. 104, no. 2, pp. 1597–1621, Nov. 2020, doi: 10.1007/s11069-020-04235-x.

[4] C. H. Smith, B. Skahill, and D. A. Margo, "Nonstationary Flood Frequency Analysis for Urban Watersheds Using Open-Source Bayesian Software: Contrasting Case Studies from Texas," *Water*, vol. 18, no. 5, p. 636, Jan. 2026, doi: 10.3390/w18050636.